

Evaluating User and Attribute Privacy in Speech via Membership Inference Attacks

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Motivation

Average Metrics Hide Real Privacy Risk

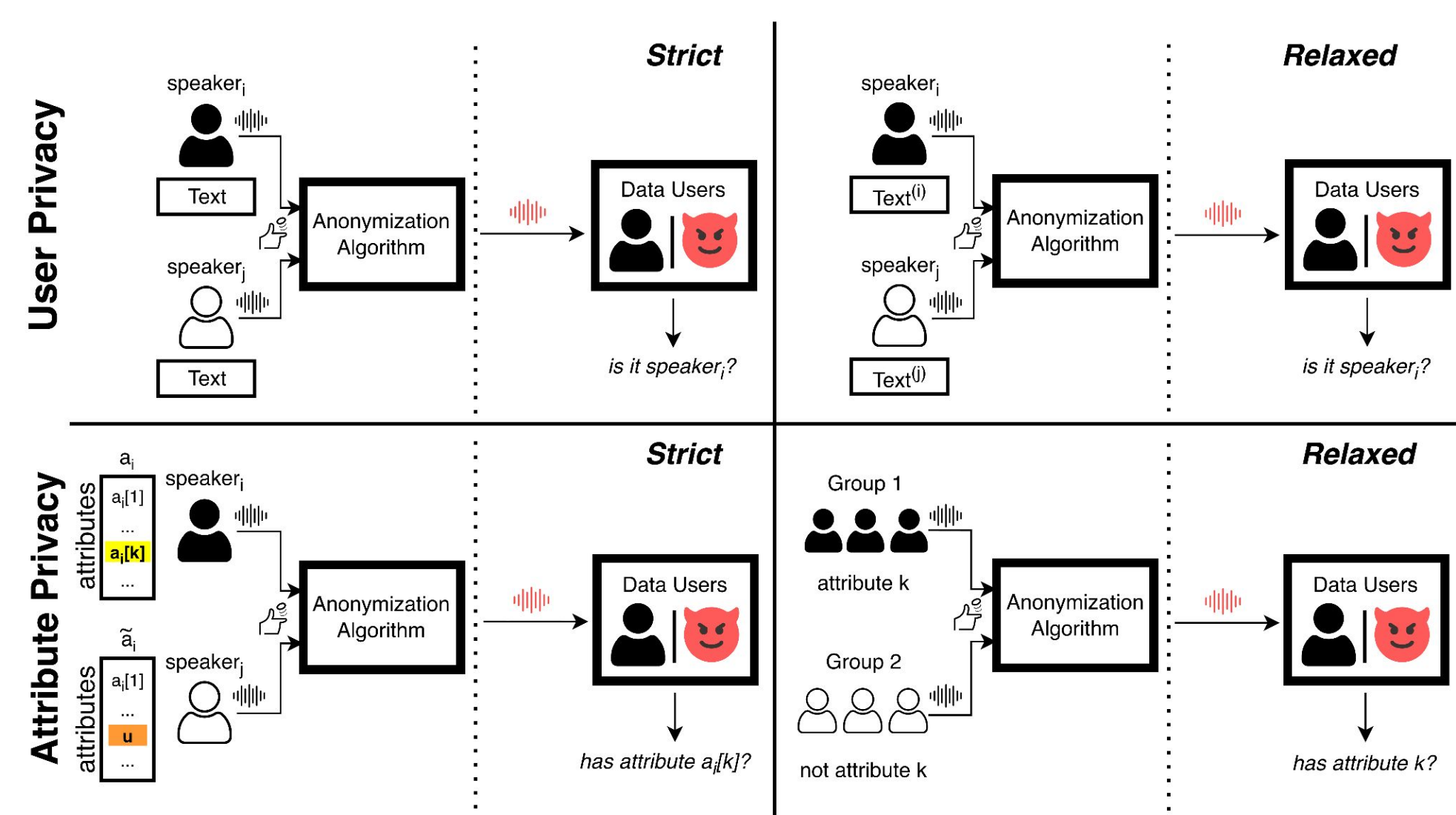
- Speech anonymization is commonly evaluated using EER.
- EER is an average-case metric and fails to capture high confidence attacks.
- In privacy, even a few successful high-precision identifications constitute a privacy breach.
- Once flagged with high confidence, a user loses plausible deniability.

What Does Speech Leaks?

Privacy Beyond Identity

- Voice encodes not only identity but also sensitive attributes.
 - Gender, accent, age, emotion.
- Existing evaluations focus almost exclusively on speaker re-identification.
- Attribute-level leakage remains largely unmeasured.

VoxGuard Framework



Differential Privacy Inspired Evaluation via MIA

- **User Privacy:**
 - Neighboring datasets differ in speaker identity
 - Evaluated as membership inference on anonymized outputs
- **Attribute Privacy:**
 - Neighboring datasets differ in attribute values
 - Captures feature-level leakage beyond identity.
- Both are quantified using DP-style bounds (ϵ, δ)

Threat Models

Informed Adversaries

- Full access to the anonymization system.
- Can generate anonymized data independently.

We evaluate:

- Pretrained-Informed and Fine-tuned Informed Attackers
- Enrollment size $L \in \{1, 3, 30, 60, 200\}$

Attribute Privacy Evaluation

Setup

- Attributes: Gender, Accent (North America, British, South Asia)
- Attribute embeddings extracted using Vox-Profile

Results

- Cosine similarity failed to reveal trends
- Switched to transparent learned attacks
 - Trained with in-distribution samples provides near perfect attribute distinguishability.
 - Trained with out-of-distribution samples favors pretrained model embeddings.

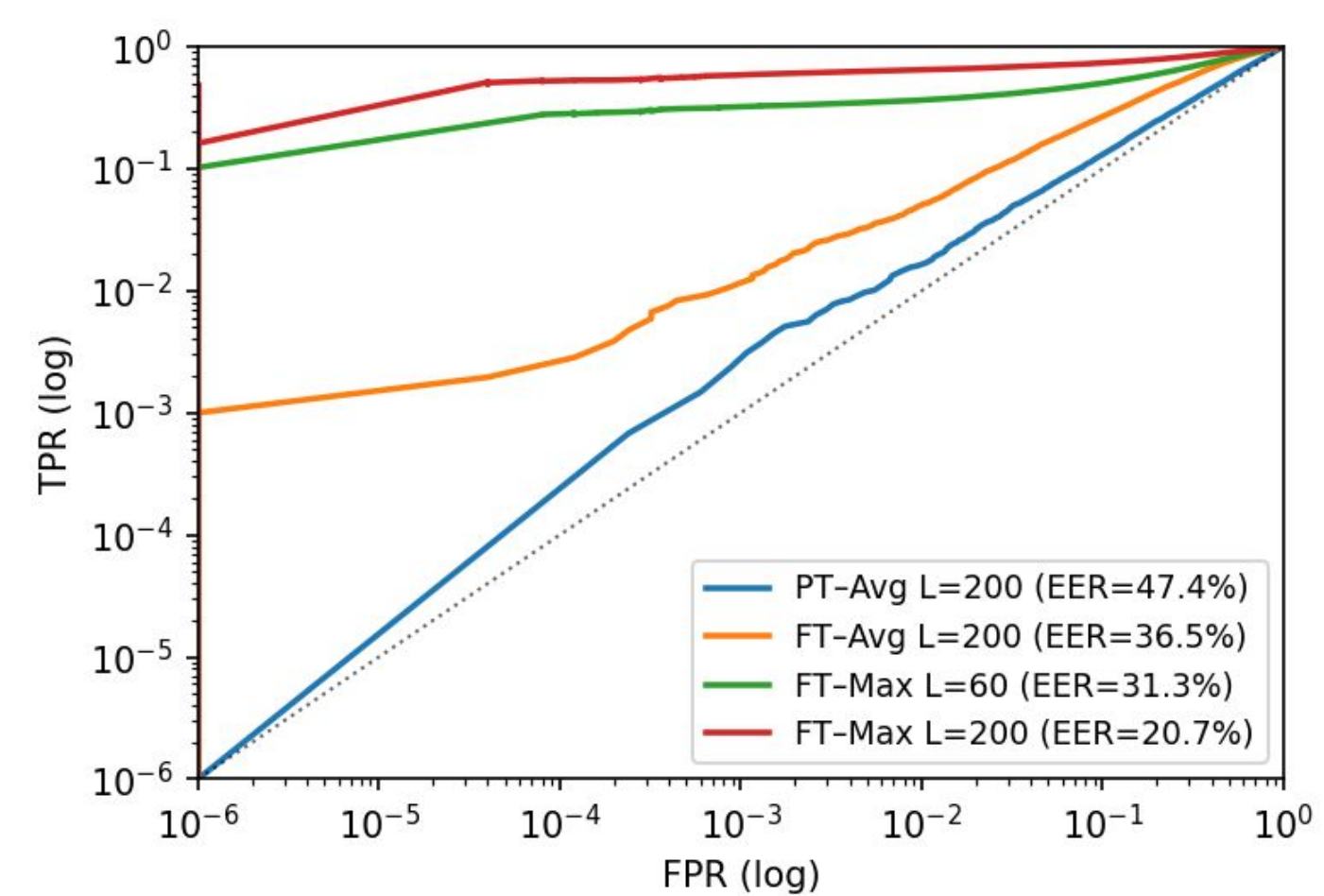
User Privacy Evaluation

Setup

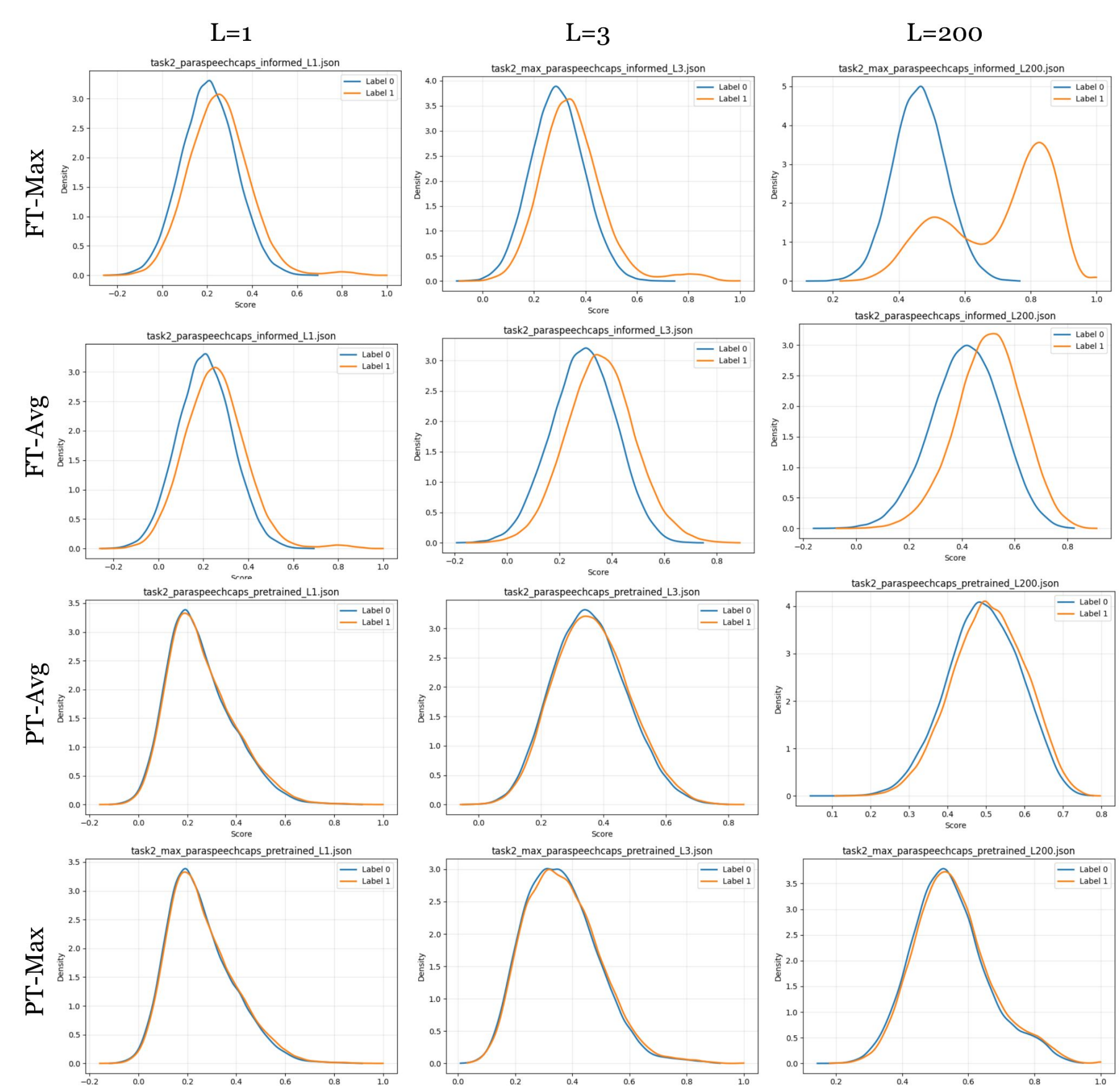
- Attribute embeddings extracted using ECAPA-TDNN
- Scoring Strategies: Average and Max Cosine Similarity

Results

- Models with similar EERs can differ by orders of magnitude at low FPR.
- Max similarity preserves high-confidence matches.
- Fine-tuning amplifies leakage.
- Larger enrollment size $L \rightarrow$ stronger attacks



Same vs Different Speaker Distributions



As enrollment size L increases:

- A second high-score mode emerges in same-speaker distribution.
- This mode is driven by linguistic overlap.
- Effect is amplified by fine-tuned embeddings.

Takeaways

- EER severely underestimates privacy risks
- Low-FPR evaluations exposes worst-case leakage
- Attribute privacy is largely unprotected
- Linguistic content can leak identity after anonymization
- Attacker adaptation and scoring strategy matters



<https://arxiv.org/abs/2509.18413>